

Research Methodology

Journal Quality and Metrics

Equipping You with Research Depth and
Industry Skills

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Journals vs. Conferences

- **Journals**

- Long, detailed, complete studies.
- **Peer review process** is rigorous (months–years).
- Higher credibility (especially if **WoS/Scopus indexed**).
- Can be *open access* (free to read, paid to publish) or subscription-based.

- **Conferences**

- Short papers, often early-stage or work-in-progress.
- Fast review cycle (weeks–months).
- Good for **networking & feedback**.
- Often the place to present **latest trends** before journal publication.



Journal Paid vs. Free

- **Paid Journals (Open Access)**
- Authors pay **Article Processing Charges (APCs)** to publish.
- Papers are then **free to read** for everyone.
- Increases **visibility & citations**.
- Examples: IEEE Access, MDPI journals.
- **2. Free-to-Publish Journals**
- Authors don't pay fees.
- Readers (universities/libraries) pay via **subscriptions**.
- Examples: Many Elsevier, Springer journals.
- **3. Predatory Journals (Warning)**
- Ask for **high fees** but provide **no real peer review**.
- Often promise **fast publication**.
- Not indexed in **WoS/Scopus** → low or no academic value.
- **Tip for Students:** Always check if a journal is indexed in **WoS/Scopus** before submitting.

What is Peer Review?

- Peer review is the process where **other experts in the same field** (peers) evaluate a research paper before it is published in a journal/conference.
- Ensures **quality, accuracy, and novelty**.
- Acts as a **filter** against poor or fake research.

Peer Review Process (Step by Step)

- **Manuscript Submission**

- Author submits paper to a journal.
- Editor checks **scope, formatting, plagiarism**.

- **Editor's Initial Screening**

- If paper is too weak → **desk rejection**.
- If promising → sent to reviewers.

- **Reviewer Selection**

- Editor invites **2–4 experts** in the same field.
- Reviewers are usually anonymous.

- **Review Process**

- Reviewers check:
 - a. Novelty of the research.
 - b. Correctness of methods/results.

- c. Clarity, references, contribution.

- They recommend:

- a. **Accept, Minor Revision, Major Revision, or Reject.**

- **Author Revision**

- Authors respond to reviewers' comments.
- Submit revised manuscript + **response letter**.

- **Final Decision by Editor**

- Based on reviewers' feedback.
- Decision: **Accept / Revise again / Reject**.

- **Publication**

- Accepted paper goes to **production** → published online/print.



Core Metrics

- **Journals:** IF, CiteScore, SJR, SNIP, Quartiles.
- **Authors:** h-index, i10-index, citation count.



Impact Factor (IF)

- A measure of how often, on average, articles in a journal are cited in a year.
- Calculated using **Web of Science (Journal Citation Reports – JCR)**.

- Journal X published **200 papers** in 2022–2023.
- In 2024, those papers received **1,000 citations**.

- $IF(2024) = 1000 \div 200 = 5.0$

- **Interpretation:**

- $IF = 5 \rightarrow$ each paper cited about 5 times/year.
- Higher IF = generally more influential journal.

- **Limitations:**

- Varies by field (medicine journals have higher IF than CS).
- Cannot compare journals across disciplines.
- Can be influenced by a few highly cited papers.

$$IF = \frac{\text{Citations in year X to papers published in (X-1, X-2)}}{\text{Total papers published in (X-1, X-2)}}$$

Citation Impact

- Journal Impact Factor: 48.5 (2024)
- 5-year Journal Impact Factor: 55.0 (2024)
- Immediacy Index: 11.8 (2024)
- Eigenfactor® Score: 0.97161 (2024)
- Article Influence Score: 23.370 (2024)
- Source Normalized Impact per Paper (SNIP): 10.161
- SCImago Journal Rank (SJR): 18.288 (2024)

<https://www.nature.com/nature/journal-impact>

Bibliometrics

IEEE Access was included in the 2024 Journal Citation Reports™ (JCR)† and received:

- **a Journal Impact Factor™ of 3.6** *Impact factor = citations in 2024 to articles published in 2022 and 2023/articles published in 2022 and 2023
- **an Eigenfactor® Score of 0.34597** *Measured by counting the total number of citations a journal receives over a five-year period
- **an Article Influence® Score of 0.670** *This is a journal-specific Eigenfactor score. It is calculated by dividing the Eigenfactor score of the journal by the number of articles published. It tells the average impact of an article published in a journal from a five-year period after the publication of the article.

<https://open.ieee.org/publishing-options/ieee-access/>

Quartile Rankings (Q1–Q4)

- Journals ranked into **four groups (quartiles)** based on impact within their field.
- Provided by **Scopus (SJR – SCImago Journal Rank)**.
- **How it works:**
 - Journals are ranked within a subject category.
 - Top 25% → **Q1 (highest prestige)**
 - Next 25% → **Q2**
 - Next 25% → **Q3**
 - Bottom 25% → **Q4 (lowest impact)**
- **Example (AI/ML field):**
 - *IEEE Transactions on Neural Networks & Learning Systems* → **Q1**
 - *Neural Processing Letters* → **Q2**
 - *International Journal of Computers & Applications* → **Q4**
- **Why useful:**
 - Easy to compare journals within the *same field*.
 - Helps students target **Q1/Q2 journals** for strong recognition.
- **Limitations:**
 - Quartiles differ across fields (a Q1 in Education ≠ Q1 in AI).
 - Some journals may be indexed in **Scopus but not WoS**.



SCI vs. SCIE vs. ESCI

- **SCI – Science Citation Index**
- Original index by **Clarivate (Web of Science)**.
- Covers **highly selective, top-tier journals** in science & technology.
- Smaller in size, very prestigious.
- Used in early days for **Impact Factor calculations**
- **SCIE – Science Citation Index Expanded**
- Broader version of SCI (also by Clarivate).
- Covers **more journals, more subject areas**.
- Currently the **main database used for Impact Factor (IF)**.
- When someone says “SCI journal,” they usually mean **SCIE-indexed journal** today.
- **ESCI – Emerging Sources Citation Index**
- Introduced in 2015 by Clarivate.
- Covers **new or regional journals** being evaluated for inclusion in SCIE/SSCI.
- Not given Impact Factor initially, but provides visibility.
- Some universities **don’t count ESCI journals** for promotions or funding.



Summary (Comparison Table)

Index	Full Form	Coverage	Impact Factor?	Prestige
SCI	Science Citation Index	Small, highly selective	Yes	Very High
SCIE	Science Citation Index Expanded	Larger, current standard	Yes	High
ESCI	Emerging Sources Citation Index	New/emerging journals	No (but visible)	Medium–Low



Where to Search for Papers (Start Research)

- **1. Free & General Search**
 - **Google Scholar** → broad coverage, free, easy to use.
 - Use for *initial exploration* and finding related works.
- **2. Subject-Specific Databases**
 - **IEEE Xplore** → computer science, AI, engineering.
 - **ACM Digital Library** → computing & software.
 - **PubMed** → medicine, biology.
 - **arXiv** → preprints in AI, physics, math.
- **3. Quality Databases (Subscription / University Access)**
 - **Web of Science (WoS – Clarivate)** → selective, trusted, Impact Factor journals.
 - **Scopus (Elsevier)** → broad, includes quartile rankings (Q1–Q4).
- **4. AI-Powered Tools**
 - **Semantic Scholar** → AI-based recommendations.
 - **Research Rabbit** → visual maps of topics & authors.
 - **Connected Papers** → explore relationships between papers.
- **Tips for Students:**
 - Start broad (Google Scholar), then go **narrow & high-quality** (WoS/Scopus).
 - Always check if a paper is from a **reputed journal or conference**.
 - Use **keywords + filters (year, author, field)** for precise results.



How to Use Keywords & Filters in Paper Search

- **1. Keywords (Search Terms)**
- Use **specific words/phrases** related to your topic.
- Example: Instead of “AI”, use “*deep learning for medical image segmentation*”.
- **2. Boolean Operators**
- **AND** → narrows search (must include both).
 - Example: “*deep learning AND liver tumor*”
- **OR** → broadens search (either term).
 - Example: “*cyberbullying OR hate speech*”
- **NOT** → excludes terms.
 - Example: “*neural networks NOT reinforcement*”
- **“Quotes”** → exact phrase.
 - Example: “*author name disambiguation*”
- * (wildcard) → variations.
 - Example: “*comput**” → computer, computing, computation.
- **3. Filters (Refining Results)**
- **By Year:** Select recent years (e.g., 2020–2025) for up-to-date research.
- **By Author:** Search within works of a specific researcher.
- **By Field/Subject:** Limit to AI, Medicine, Social Sciences, etc.
- **By Publication Type:** Journals vs. Conferences.

